# A SHARP ASYMPTOTIC THRESHOLD FOR TERMINAL EXTRAPOLATION OF GRADIENT DESCENT

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Abstract. We consider the gradient norm at the extrapolated output $y = x_0 + c(x_N - x_0)$ of $N$ iterations of gradient descent with normalized constant step size $h \in (0, 1]$ on smooth convex functions. The one-dimensional Huber example gives the lower bound $LD/(1 + Nhc)$ when the initial distance to a minimizer is at most $D$ and the gradient is $L$-Lipschitz. We determine the asymptotic endpoint of the maximal initial interval of extrapolation factors on which this lower bound equals the worst-case gradient norm. For each fixed $h \in (0, 1]$, the threshold satisfies

$$c_{\mathrm{crit}}(N, h) - 1 \sim \frac{1}{hN\sqrt{\log N}}.$$

The proof combines a smooth-convex interpolation certificate obtained from a constrained quadratic program, a tridiagonal energy estimate with matching endpoint profiles, and a two-dimensional smooth convex construction giving a matching asymptotic upper bound on the threshold. The logarithmic factor corrects the order conjectured by Luner and Grimmer.

## 1. Introduction

After $N$ gradient descent steps, terminal extrapolation replaces the final iterate by $y_c = x_0 + c(x_N - x_0)$, with $c \geq 1$. This output uses only the initial and final iterates and requires no additional gradient evaluations. Luner and Grimmer [2] established sharp worst-case function-value guarantees for terminal extrapolation and studied the corresponding gradient-norm problem numerically. Conjecture 5.1 in the arXiv version of their work proposes that, for fixed normalized step size $h \in (0, 1]$, the critical excess extrapolation factor is $\Theta(N^{-1})$.

We determine the asymptotic endpoint of the initial interval on which the Huber gradient-norm bound $LD/(1 + Nhc)$ is exact. For every fixed $h \in (0, 1]$, we prove

$$c_{\mathrm{crit}}(N, h) - 1 \sim \frac{1}{hN\sqrt{\log N}}.$$

The excess is therefore $o(N^{-1})$, correcting the order proposed in the conjecture. The result combines equality throughout the lower interval with a two-dimensional smooth convex example giving strict failure at the matching upper scale.

The proof uses performance estimation [1] and smooth-convex interpolation [3]. A nonnegative directed flow specifies the coefficients of an interpolation certificate. We construct this flow from the Karush–Kuhn–Tucker conditions of a constrained quadratic program. A tail-sum transformation expresses its objective through a positive tridiagonal energy and an explicit negative correction. Comparing the transformed trial vector with a reciprocal model reduces the leading energy term to two harmonic sums and identifies the critical scale. A negative trial direction supplies the finite interpolation data for the planar counterexample, which is realized by convex conjugacy.

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Sections 3–5 develop the certificate, analyze the quadratic program, and construct the counterexample. Section 6 combines these results to prove the theorem. The detailed flow construction and energy estimates appear in Appendices A and B.

## 2. Setting and main theorem

For $L > 0$ and $d \geq 1$, let $\mathcal{F}_L(\mathbb{R}^d)$ be the class of convex differentiable functions $f : \mathbb{R}^d \to \mathbb{R}$ having an $L$-Lipschitz gradient and attaining their minimum. Let $x_\star$ be a minimizer with $\|x_0 - x_\star\| \leq D$, where $D > 0$. For $N \geq 1$, $0 < h \leq 1$, and $c \geq 1$, consider

$$x_{i+1} = x_i - \frac{h}{L}\nabla f(x_i) \quad (0 \leq i < N), \qquad y_c = x_0 + c(x_N - x_0). \tag{2.1}$$

The dimension-independent worst-case gradient norm is

$$P_{N,h}(c; L, D) = \sup\left\{ \|\nabla f(y_c)\| : \begin{array}{l} d \geq 1,\ f \in \mathcal{F}_L(\mathbb{R}^d), \\ x_\star \in \operatorname{argmin} f,\ x_0 \in \mathbb{R}^d,\ \|x_0 - x_\star\| \leq D \end{array} \right\}. \tag{2.2}$$

Define the endpoint of the initial interval of exactness by

$$c_{\mathrm{crit}}(N, h) = \sup\left\{ C \geq 1 :\ P_{N,h}(c; L, D) = \frac{LD}{1 + Nhc} \quad \text{for every } c \in [1, C] \right\}. \tag{2.3}$$

Translation and scaling show that $c_{\mathrm{crit}}$ is independent of $L, D$. Throughout, log denotes the natural logarithm.

**Theorem 2.1.** *For every fixed $h \in (0, 1]$,*

$$\lim_{N\to\infty} hN\sqrt{\log N}\,(c_{\mathrm{crit}}(N, h) - 1) = 1. \tag{2.4}$$

*More precisely, for every $h \in (0, 1]$ and $\varepsilon \in (0, 1)$, there exists $N_0 = N_0(h, \varepsilon) \geq 2$ such that, for all integers $N \geq N_0$ and all $L, D > 0$,*

$$P_{N,h}(c; L, D) = \frac{LD}{1 + Nhc} \quad \textit{for } 1 \leq c \leq 1 + \frac{1 - \varepsilon}{hN\sqrt{\log N}}, \tag{2.5}$$

*whereas, at $c_+ = 1 + (1 + \varepsilon)/(hN\sqrt{\log N})$,*

$$P_{N,h}(c_+; L, D) > \frac{LD}{1 + Nhc_+}. \tag{2.6}$$

*For each such $N, L, D$, there exist $f \in \mathcal{F}_L(\mathbb{R}^2)$, a minimizer $x_\star$, and an initial point with $\|x_0 - x_\star\| = D$ for which $\|\nabla f(y_{c_+})\| > LD/(1 + Nhc_+)$.*

Henceforth we normalize $L = D = 1$, $x_\star = 0$, and $f(x_\star) = 0$. The general result follows from the transformation $f(x) = f(x_\star) + LD^2\widetilde{f}((x - x_\star)/D)$. The worst-case value is unchanged if the radius constraint is imposed with equality: for an instance of radius $r < D$, extend $f$ to $\mathbb{R}^{d+1}$ by making it independent of the last coordinate and take that initial coordinate to be $\sqrt{D^2 - r^2}$. The trajectory and output gradients in the original coordinates are unchanged.

**Lemma 2.2** (Huber lower bound)**.** *For every $N, h, c$ in the stated ranges, $P_{N,h}(c; 1, 1) \geq (1 + Nhc)^{-1}$.*

*Proof.* Put $\eta = (1 + Nhc)^{-1}$ and take $x_0 = 1$ and

$$f_\eta(x) = \begin{cases} \frac{1}{2}x^2, & |x| \leq \eta, \\ \eta|x| - \frac{1}{2}\eta^2, & |x| > \eta. \end{cases}$$

All gradients queried by the algorithm equal $\eta$, and $x_i = 1 - ih\eta$ for $0 \leq i \leq N$. Thus $y_c = \eta$ and $|f'_\eta(y_c)| = \eta$. □

The supremum in (2.2) is finite. Cocoercivity gives $\|x_i\| \le \|x_0\| \le 1$ for $0 < h \le 1$, whence $\|y_c\| \le 2c-1$ and $\|\nabla f(y_c)\| \le 2c-1$. Lemma 4.2 below gives equality on $[1, 1+\delta]$ for some $\delta > 0$. For $f(x) = x^2/2$ and $x_0 = 1$, the output gradient has magnitude $|1 - c[1-(1-h)^N]|$, which exceeds $(1+Nhc)^{-1}$ for all sufficiently large $c$. Thus the set in (2.3) is nonempty and bounded, and $1 < c_{\mathrm{crit}}(N,h) < \infty$.

## 3. Interpolation certificates and an auxiliary quadratic program

In the normalized setting write $g_i = \nabla f(x_i)$ and $q = \nabla f(y_c)$. We index the queried gradients by $i = 0, \dots, N-1$, the output by $o$, and the minimizer by $\star$. In vector and graph notation, node $N$ denotes the output $o$.

We seek an upper bound by combining interpolation inequalities with nonnegative weights. The function values will cancel, leaving a quadratic form in the gradient differences $g_i - q$. Choosing the weights to make this form positive semidefinite gives the required gradient bound.

The smooth-convex interpolation inequalities are

$$I_{ij} := f_i - f_j - \langle g_j, x_i - x_j\rangle - \frac{1}{2}\|g_i - g_j\|^2 \ge 0 \qquad (i,j \in \{\star, 0, \dots, N-1, o\}). \tag{3.1}$$

Indeed, let $\varphi(x) = f(x) - \langle g_j, x\rangle$. Then $x_j$ minimizes $\varphi$ and the descent lemma gives

$$\varphi(x_j) \le \varphi(x_i - \nabla\varphi(x_i)) \le \varphi(x_i) - \tfrac{1}{2}\|g_i - g_j\|^2,$$

which is (3.1).

Set

$$S = 1 + Nhc, \qquad K_i = h(Nc - i) > 0, \qquad T = hc\mathbf{1}\mathbf{1}^\top - hL_< - I, \tag{3.2}$$

where $L_<$ is the strictly lower triangular all-ones matrix. Consider nonnegative weights $F_{ij}$, $0 \le i < j \le N$, satisfying

$$\sum_{j>i} F_{ij} - \sum_{k<i} F_{ki} = 1 \qquad (0 \le i < N). \tag{3.3}$$

Write $e_i$ for the standard basis of $\mathbb{R}^N$, set $e_N = 0$, and denote by $T_i$ the $i$th row of $T$. Define

$$\lambda_{ij} = \frac{2hcS}{K_i} F_{ij}, \qquad b_i = \sum_{j>i} \lambda_{ij} - \sum_{k<i} \lambda_{ki} - 2hc, \tag{3.4}$$

$$M(F) = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS \sum_{i<j} \frac{F_{ij}}{K_i}\left[(e_i - e_j)T_i + T_i^\top (e_i - e_j)^\top\right]. \tag{3.5}$$

The coefficients $b_i$ are positive, since

$$b_i = 2hc\left[\frac{S}{K_i} - 1 + S\sum_{k<i} F_{ki}\left(\frac{1}{K_i} - \frac{1}{K_k}\right)\right] > 0. \tag{3.6}$$

**Lemma 3.1** (Forward-flow certificate)**.** *For every flow satisfying (3.3), with $v_i = g_i - q$, $v_N = 0$, and $z = y_c - q$, one has*

$$\begin{aligned}\|x_0\|^2 - S^2\|q\|^2 = \|z\|^2 + \sum_{i,j<N} M(F)_{ij}\langle v_i, v_j\rangle + 2hc\sum_{i<N} I_{\star i} + 2I_{\star o} + 2SI_{o\star} \\ + \sum_{i<j} \lambda_{ij} I_{ij} + \sum_{i<N} b_i I_{oi}.\end{aligned} \tag{3.7}$$

*In particular, $M(F) \succeq 0$ implies $P_{N,h}(c;1,1) = (1+Nhc)^{-1}$.*

*Proof.* Identity (3.7) is proved by expansion in Lemma A.1. Every term on the right-hand side is nonnegative when $M(F) \succeq 0$, using (3.1), (3.6), and $F \geq 0$. Together with $\|x_0\| \leq 1$, this gives the upper bound. Lemma 2.2 gives equality. □

The adjacent-chain flow $F_{i,i+1} = i+1$ defines

$$M_0 = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS(D^\top WT + T^\top WD), \quad W = \operatorname{diag}\left(\frac{i+1}{K_i}\right)_{i=0}^{N-1}, \tag{3.8}$$

where $(D\beta)_i = \beta_i - \beta_{i+1}$ and $\beta_N = 0$. Write $R = \operatorname{diag}(K_i^{-1})T$, $\tau = R\beta$, and $(A\beta)_i = \tau_i - \tau_{i+1}$ for $i < N-1$. Let

$$\mathcal{K} = \{\beta \in \mathbb{R}^N : \mathbf{1}^\top\beta = 1,\ D\beta \geq 0,\ A\beta \geq 0\}, \qquad \Phi_{N,h}(c) = \min_{\beta\in\mathcal{K}} \beta^\top M_0\beta. \tag{3.9}$$

Thus there are $N$ monotonicity conditions on $\beta$ (including $\beta_{N-1} \geq 0$), and $N-1$ on $\tau$. For $\beta \in \mathcal{K}$, write $\delta_i = \beta_i - \beta_{i+1}$ and

$$Q(\beta) = \beta^\top M_0\beta = h^2c^2 - 2hcS\sum_{i=0}^{N-1}(i+1)\delta_i\tau_i. \tag{3.10}$$

The quadratic program selects the vector used to construct the certificate. Its KKT multipliers determine a flow for which $M(F)\beta = \Phi_{N,h}(c)\mathbf{1}$; strict positivity of $\beta$ and the signs of the off-diagonal entries then give a sum-of-squares decomposition. The energy representation in the next section establishes existence and uniqueness of the optimizer. Appendix A carries out the flow construction.

**Proposition 3.2** (Quadratic-program sufficiency)**.** *If $N \geq 2$, $0 < h < 1$, $Nh^2 \geq 1$, $c \geq 1$, and $\Phi_{N,h}(c) \geq 0$, then*

$$P_{N,h}(c;1,1) = \frac{1}{1+Nhc}. \tag{3.11}$$

*Proof.* Proposition A.5 constructs a flow satisfying (3.3) and $M(F) \succeq 0$ from the optimizer of (3.9). Lemma 3.1 applies. □

## 4. A tridiagonal energy and the transition law

Put

$$t = h^{-1} \geq 1, \quad a = N(c-1) \geq 0, \quad d_i = N+a-i, \quad \mathfrak{F} = N+a+t, \quad \zeta_i = \mathfrak{F}-i = d_i+t. \tag{4.1}$$

For $p \in \mathbb{R}^N$ with $p_N = 0$, define $w_i = \sum_{j=i}^{N-1} p_j$, $z_i = w_i/\zeta_i$, $z_N = 0$, and $u_i = z_i - z_{i+1}$ for $0 \leq i < N$. Set $u_N = 0$. The map $p \mapsto u$ is invertible, and

$$p_i = \zeta_i u_i + \sum_{j>i} u_j, \qquad \sum_i p_i = \mathfrak{F}\sum_i u_i, \qquad p_i - p_{i+1} = \zeta_i u_i - (\zeta_i - 2)u_{i+1}. \tag{4.2}$$

Set

$$\mathcal{E}_{a,t}(p) = \left(\sum_i p_i\right)^2 - 2\mathfrak{F}\sum_{i=0}^{N-1}(i+1)(p_i - p_{i+1})\frac{w_i - tp_i}{d_i}. \tag{4.3}$$

**Lemma 4.1** (Tridiagonal reduction)**.** *For every $a \geq 0, t \geq 1$,*

$$\mathcal{E}_{a,t}(p) = \mathfrak{F}^2 u^\top Eu, \tag{4.4}$$

*where $E$ is symmetric tridiagonal with*

$$E_{ii} = \frac{2t(i+1)\zeta_i(\zeta_i - 1)}{d_i\mathfrak{F}} - 1, \qquad E_{i,i+1} = -\frac{(t-1)(i+1)\zeta_i(\zeta_i - 2)}{d_i\mathfrak{F}}. \tag{4.5}$$

*Moreover,*

$$E \succeq \operatorname{diag}(e_0^0, \dots, e_{N-1}^0) \succ 0, \qquad e_i^0 = 2i + 1 - \frac{2i(i+1)}{N+1}. \tag{4.6}$$

*Proof.* The identities are established in Lemma B.1. In particular, $\mathcal{E}_{a,t}(p) > 0$ for $p \neq 0$. □

If $\mathbf{1}^\top v = 0$, substitution into (3.8) gives

$$\frac{v^\top M_0 v}{h^2 c} = \mathcal{E}_{a,t}(v) > 0 \qquad (v \neq 0). \tag{4.7}$$

**Lemma 4.2** (Initial interval of exactness)**.** *For every $N \geq 1$ and $h \in (0, 1]$, there exists $\delta > 0$ such that $P_{N,h}(c; 1, 1) = (1 + Nhc)^{-1}$ for all $c \in [1, 1 + \delta]$.*

*Proof.* At $c = 1$, substitution into (3.8) gives $v^\top M_0 v = h^2 \mathcal{E}_{0,1/h}(v)$ for every $v \in \mathbb{R}^N$. Lemma 4.1 therefore implies $M_0 \succ 0$ at $c = 1$. The entries of $M_0$ depend continuously on $c$, so $M_0 \succ 0$ on $[1, 1 + \delta]$ for some $\delta > 0$. The adjacent-chain flow satisfies (3.3), and Lemma 3.1 gives the result. □

The set $\mathcal{K}$ is closed and contained in the simplex, hence compact. For the uniform vector $\bar{\beta}_i = 1/N$,

$$(R\bar{\beta})_i = \frac{1}{N} - \frac{1}{NK_i}, \qquad (A\bar{\beta})_i = \frac{h}{NK_iK_{i+1}} > 0 \quad (0 \leq i < N - 1).$$

For sufficiently small $\eta > 0$, the perturbation $\widehat{\beta}_i = N^{-1} + \eta((N-1)/2 - i)$ has sum one, positive last coordinate, and positive adjacent differences. By continuity, $A\widehat{\beta} > 0$ as well. For $N = 1$, the slope constraints are empty and $\widehat{\beta}_0 = 1$ is strictly feasible. Thus $\mathcal{K}$ satisfies the relative Slater condition. Compactness gives a minimizer, and (4.7) gives its uniqueness.

Define the reference reciprocal sum

$$\mathcal{R}_N = \sum_{i=0}^{N-1} \frac{1}{e_i^0}. \tag{4.8}$$

From (4.6), (4.2), and weighted Cauchy–Schwarz,

$$\mathcal{E}_{a,t}(p) \geq \frac{1}{\mathcal{R}_N} \quad \text{whenever} \quad \sum_i p_i = 1. \tag{4.9}$$

Furthermore, $\mathcal{R}_N = \log N + O(1)$. Indeed, with $q_i = 2(i+1)(N-i)/(N+1)$,

$$e_i^0 = q_i - 1 + \frac{2(i+1)}{N+1}, \qquad \left| \frac{1}{e_i^0} - \frac{1}{q_i} \right| \leq 2 \left( \frac{1}{(i+1)^2} + \frac{1}{(N-i)^2} \right),$$

while $\sum_i q_i^{-1} = \sum_{j=1}^N j^{-1}$.

For every $\beta \in \mathcal{K}$, one has

$$\beta_i \geq B := \frac{a}{N(a+t)}. \tag{4.10}$$

To see this, set $B_i = K_i(\beta_i - \tau_i)$; the slope constraints imply $B_0 \geq N^{-1}$ and $B_{i+1} \geq (1 - h)(K_{i+1}/K_i)B_i \geq 0$, while $B_{N-1} = (1 + ha)\beta_{N-1} - ha/N$. Now let $m = t/(a+t)$ and write

$$\beta = B\mathbf{1} + mp, \qquad p \geq 0, \quad \mathbf{1}^\top p = 1. \tag{4.11}$$

Writing $\Delta p_i = p_i - p_{i+1}$, the shifted slopes and differences are

$$\tau_i = B + m\frac{w_i - tp_i}{d_i}, \qquad \delta_i = m\Delta p_i + B\mathbf{1}_{\{i=N-1\}}.$$

Consequently, with $J = \sum_i (i+1)\Delta p_i(w_i - tp_i)/d_i$,

$$\sum_i (i+1)\delta_i\tau_i = B + m^2 J + \frac{NmB(1-t)}{a+1}p_{N-1}.$$

Substitution into (3.10), together with $c - m^2 - 2\mathfrak{F}B = -a^2/(a+t)^2 - a/N$, yields

$$\frac{Q(\beta)}{h^2 c} = m^2\mathcal{E}_{a,t}(p) - \frac{a^2}{(a+t)^2} - \frac{a}{N} + \frac{2\mathfrak{F}mBN(t-1)}{a+1}p_{N-1}. \tag{4.12}$$

The final term is nonnegative; hence

$$\frac{\Phi_{N,h}(c)}{h^2 c} \geq \frac{t^2/\mathcal{R}_N - a^2}{(a+t)^2} - \frac{a}{N}. \tag{4.13}$$

The leading terms in (4.13) balance at $a^2 \sim t^2/\log N$. This suggests the scale $c - 1 \sim 1/(hN\sqrt{\log N})$. To obtain a matching upper estimate, we need a strictly feasible vector whose energy attains the leading value $1/\log N$. Fix $t \geq 1$, put $s = 2t$, and set

$$q_j^{\mathrm{H}} = \sum_{k=1}^{j} \frac{1}{k+s}, \qquad v_i = \frac{1}{i+s} + \frac{q_{N-i}^{\mathrm{H}}}{N}, \qquad H_N = \sum_{i=0}^{N-1} v_i, \qquad p_i = \frac{v_i}{H_N}. \tag{4.14}$$

**Lemma 4.3** (Matching admissible profile)**.** *For $0 \leq a \leq (2t+1)^{-1}$, the vector $\beta = B\mathbf{1} + mp$ defined by (4.11) and (4.14) satisfies $D\beta > 0$ and $A\beta > 0$. For fixed $t \geq 1$, uniformly over this range of $a$,*

$$H_N = 2\log N + O_t(1), \qquad \mathcal{E}_{a,t}(p) = \frac{1}{\log N} + o_t((\log N)^{-1}). \tag{4.15}$$

*Proof.* Strict feasibility is proved in Lemma B.2. The energy estimate is proved in Lemma B.4. □

**Theorem 4.4** (Uniform transition of the quadratic program)**.** *Fix $h \in (0,1]$ and $K > 0$. For $N \geq 2$, let $c_N(k) = 1 + k/(hN\sqrt{\log N})$. Then, as $N \to \infty$,*

$$\sup_{0\leq k\leq K} \left| \frac{\log N}{h^2 c_N(k)}\Phi_{N,h}(c_N(k)) - (1-k^2) \right| \longrightarrow 0. \tag{4.16}$$

*The same limit holds with $\Phi_{N,h}$ replaced by the objective of the strictly feasible trial vector from Lemma 4.3.*

*Proof.* With $a = N(c_N(k) - 1) = kt/\sqrt{\log N}$, the lower estimate (4.13) and $\mathcal{R}_N = \log N + O(1)$ give

$$\frac{\log N}{h^2 c_N(k)}\Phi_{N,h}(c_N(k)) \geq \frac{t^2 \log N/\mathcal{R}_N - k^2 t^2}{(a+t)^2} - \frac{a\log N}{N} = 1 - k^2 + o_{t,K}(1),$$

uniformly for $k \in [0, K]$. For large $N$, all these $a$ satisfy the condition of Lemma 4.3. Inserting its trial vector into (4.12) yields

$$\frac{Q(\beta)}{h^2 c_N(k)} = \frac{t^2}{(a+t)^2}\left(\frac{1}{\log N} + o_{t,K}((\log N)^{-1})\right) - \frac{a^2}{(a+t)^2} - \frac{a}{N} + O_{t,K}\left(\frac{a}{\log N}\right).$$

Here $p_{N-1} = O_t((N\log N)^{-1})$ controls the terminal contribution. Multiplication by $\log N$ gives $1 - k^2 + o_{t,K}(1)$ uniformly. Since $\Phi_{N,h} \leq Q(\beta)$, the two estimates prove the claim. □

## 5. Negative directions and planar counterexamples

A strictly feasible vector with negative quadratic objective yields a two-dimensional counterexample. We perturb the constant-gradient Huber data so that the output gradient has norm one while the initial radius becomes smaller than $S = 1 + Nhc$. A quadratic perturbation produces the radius improvement; a quartic correction makes the adjacent interpolation inequalities positive. The resulting finite data are then realized by a smooth convex function and rescaled to unit initial radius.

**Proposition 5.1.** *Let $N \geq 1$, $0 < h \leq 1$, and $c \geq 1$. Suppose $\beta \in \mathbb{R}^N$ satisfies*

$$\mathbf{1}^\top \beta = 1, \qquad \beta_0 > \cdots > \beta_{N-1} > 0, \qquad (R\beta)_0 > \cdots > (R\beta)_{N-1}, \qquad Q(\beta) < 0. \tag{5.1}$$

*Then $P_{N,h}(c;1,1) > (1+Nhc)^{-1}$, witnessed by a globally defined 1-smooth convex function on $\mathbb{R}^2$. The conclusion also holds if $\Phi_{N,h}(c) < 0$.*

*Proof.* Write $\tau = R\beta$, $\delta_i = \beta_i - \beta_{i+1}$ and extend $\beta_N = 0$. Define arrays $A_i, B_i$ backwards from $A_N = B_N = 0$ by

$$A_i = A_{i+1} - \delta_i \tau_i, \qquad P_i = (TA)_i, \qquad B_i = B_{i+1} + \frac{1 - (A_i - A_{i+1})P_i}{K_i}, \qquad D_i = (TB)_i. \tag{5.2}$$

First compute $A_0, \ldots, A_{N-1}$ and $P = TA$, then $B_0, \ldots, B_{N-1}$ and $D = TB$. For $\epsilon > 0$ prescribe the following points and gradients in $\mathbb{R}^2$:

$$q = y_c = (1,0), \quad g_i = (1 + A_i\epsilon^2 + B_i\epsilon^4,\ \beta_i\epsilon), \quad x_0 = q + hc\sum_{i<N} g_i, \quad x_i = x_0 - h\sum_{j<i} g_j. \tag{5.3}$$

Set $(x_o, g_o, f_o) = (q, q, 1/2)$, $(x_\star, g_\star, f_\star) = (0,0,0)$, and $f_i = \langle g_i, x_i\rangle - \|g_i\|^2/2$ at ordinary nodes. These points satisfy (2.1).

For these data, the interpolation slack (3.1) reduces to

$$I_{ij} = \langle g_i - g_j, x_i - g_i\rangle. \tag{5.4}$$

If the source is $\star$ or $o$, it vanishes. For an ordinary source $i$ we have $x_i - g_i = (K_i + P_i\epsilon^2 + D_i\epsilon^4,\ K_i\tau_i\epsilon)$. With $\Delta A = A_i - A_j$, $\Delta B = B_i - B_j$, $\Delta\beta = \beta_i - \beta_j$ and zero arrays at $j = o$, expansion gives

$$\begin{aligned} I_{ij} &= K_i(\Delta A + \tau_i\Delta\beta)\epsilon^2 + (K_i\Delta B + P_i\Delta A)\epsilon^4 \\ &\quad + (D_i\Delta A + P_i\Delta B)\epsilon^6 + D_i\Delta B\epsilon^8. \end{aligned} \tag{5.5}$$

For $j = i+1$, the quadratic coefficient is zero and the quartic coefficient equals 1. For $j \geq i+2$ the quadratic coefficient is

$$K_i\sum_{\ell=i}^{j-1}\delta_\ell(\tau_i - \tau_\ell) > 0;$$

for $j < i$ it is $K_i\sum_{\ell=j}^{i-1}\delta_\ell(\tau_\ell - \tau_i) > 0$. The remaining ordinary-to-minimizer slacks equal $K_i + O(\epsilon^2) > 0$. The diagonal slacks are zero. Since there are finitely many ordered pairs, a single sufficiently small $\epsilon > 0$ makes every remaining slack nonnegative.

Moreover $\sum_i A_i = -\sum_i (i+1)\delta_i\tau_i$, so

$$\|x_0\|^2 - S^2 = Q(\beta)\epsilon^2 + O(\epsilon^4) < 0. \tag{5.6}$$

To realize the data, define

$$\psi(u) = \tfrac{1}{2}\|u\|^2 + \max_{j\in\{\star,0,\ldots,N-1,o\}}\langle x_j - g_j, u - g_j\rangle, \qquad f = \psi^*. \tag{5.7}$$

The function $\psi$ is closed, finite and 1-strongly convex. For each $x \in \mathbb{R}^2$, the function $u \mapsto \psi(u) - \langle x, u\rangle$ is coercive and strongly convex, so it has a unique minimizer $u(x)$, characterized by $x \in \partial\psi(u(x))$. Hence $f = \psi^*$ is finite and differentiable, with $\nabla f(x) = u(x)$. Strong monotonicity of $\partial\psi$ gives

$$\|u(x) - u(y)\|^2 \le \langle x - y, u(x) - u(y)\rangle \le \|x - y\|\,\|u(x) - u(y)\|,$$

so $\nabla f$ is 1-Lipschitz. Convexity follows from the definition of the conjugate. At $u = g_i$, the affine term indexed by $j$ equals $\langle x_j - g_j, g_i - g_j\rangle = -I_{ji} \le 0$, while the term indexed by $i$ is zero. Thus $\psi(g_i) = \|g_i\|^2/2$ and $x_i \in \partial\psi(g_i)$. Fenchel equality gives $f(x_i) = \langle x_i, g_i\rangle - \psi(g_i) = f_i$ and $\nabla f(x_i) = g_i$. In particular, $\nabla f(0) = 0$, so 0 minimizes $f$.

Let $R_* = \|x_0\|$. By (5.6), $0 < R_* < S$. Rescale to $\widetilde{f}(x) = f(R_* x)/R_*^2$ and initial point $x_0/R_*$. This preserves the step size and the output relation, and produces $\|\nabla\widetilde{f}(y_c/R_*)\| = 1/R_* > 1/S$.

Finally, if $\Phi_{N,h}(c) < 0$, mix the optimizer of (3.9) with a strictly feasible point of $\mathcal{K}$. A sufficiently small mixing weight preserves negativity of the objective and makes both orders strict, so the preceding construction applies. □

## 6. Proof of the main theorem

*Proof of Theorem 2.1.* Fix $0 < \varepsilon < 1$. First take $0 < h < 1$. By Theorem 4.4, for all sufficiently large $N$, $\Phi_{N,h}(c_N(k)) > 0$ for every $0 \le k \le 1 - \varepsilon$. Since $h$ is fixed, $Nh^2 \ge 1$ eventually. Proposition 3.2 and Lemma 2.2 give (2.5) on the full indicated interval.

At $k = 1 + \varepsilon$, the strictly feasible trial vector in Lemma 4.3 has negative objective for all sufficiently large $N$. Proposition 5.1 gives (2.6). Every $C \ge c_+$ includes this failure point and is therefore excluded from the set in (2.3). Consequently

$$\frac{1 - \varepsilon}{hN\sqrt{\log N}} \le c_{\mathrm{crit}}(N, h) - 1 \le \frac{1 + \varepsilon}{hN\sqrt{\log N}}.$$

For $h = 1$, the negative trial vector and Proposition 5.1 remain valid. To obtain the lower interval, fix large $N$ and $c$ in the interval in (2.5) with $h = 1$. The uniform estimate for the right-hand side of (4.13) in the proof of Theorem 4.4 makes this bound strictly positive throughout the indicated interval for large $N$. For fixed $N, c$, that right-hand side is continuous in $h$ at 1. Choose $h_m \uparrow 1$ with $h_m < 1$, $Nh_m^2 \ge 1$, and $\Phi_{N,h_m}(c) > 0$. The sufficient bound applies for each $h_m$. For each fixed admissible function and initial point, its finite-step trajectory, extrapolated output and gradient are continuous in $h$. Passing to the limit yields $\|\nabla f(y_c)\| \le (1 + Nc)^{-1}$. Taking the supremum and invoking Lemma 2.2 gives (2.5) at $h = 1$. Letting $\varepsilon \downarrow 0$ proves (2.4) in all cases, and rescaling restores $L, D$. □

## 7. Concluding remarks

Theorem 2.1 identifies the leading asymptotic term of the critical extrapolation factor. Further questions include the finite-$N$ worst-case performance curve, the optimal extrapolation factor, a second-order threshold expansion, and joint limits in which $h$ decreases with $N$.

## Appendix A. Construction of the interpolation certificate

### A.1. The forward-flow identity.

**Lemma A.1.** *The forward-flow identity* (3.7) *holds for any flow satisfying* (3.3).

*Proof.* Let $a_i = f_i - \langle g_i, x_i\rangle + \|g_i\|^2/2$. Then $I_{ij} = a_i - a_j + \langle g_i - g_j, x_i - g_i\rangle$. In the weighted interpolation sum on the right of (3.7), the coefficient of each ordinary $a_i$ is $-2hc + \sum_{j>i}\lambda_{ij} - \sum_{k<i}\lambda_{ki} - b_i = 0$. Those of $a_o, a_\star$ vanish by (3.3) and $S = 1 + Nhc$. Furthermore,

$$x_0 = Sq + z + hc\sum_i v_i, \qquad x_i - g_i = K_i q + z + (Tv)_i.$$

Flow conservation implies $\sum_{i<j}\lambda_{ij}K_i(v_i - v_j) = 2hcS\sum_i v_i$. The weighted slacks therefore reduce to

$$2S\langle q, z\rangle + 2hc\langle\sum_i v_i, z\rangle + 2hcS\langle q, \sum_i v_i\rangle + \sum_{i<j}\lambda_{ij}\langle v_i - v_j, (Tv)_i\rangle.$$

Expanding the squared norms and using (3.5) gives the remaining quadratic term in (3.7). □

A.2. **The optimizer and its order structure.** We now prove Proposition 3.2. The construction has three steps: establish strict order of the optimizer, use its KKT multipliers to define a nonnegative flow, and prove positivity of the resulting certificate matrix. Write $r = 1 - h \in (0, 1)$, and let $\beta$ be the unique optimizer of (3.9). The KKT equations have multipliers $\mu \in \mathbb{R}_+^{N-1}$ and $\nu \in \mathbb{R}_+^N$ satisfying

$$2M_0\beta = 2\Phi_{N,h}(c)\mathbf{1} + A^\top\mu + D^\top\nu, \qquad \mu_i(\tau_i - \tau_{i+1}) = 0, \quad \nu_i\delta_i = 0. \tag{A.1}$$

The equality multiplier is $2\Phi_{N,h}(c)$ by multiplication by $\beta^\top$. Define $B_i = K_i(\beta_i - \tau_i)$. A direct calculation gives

$$B_0 = S\beta_0 - hc \geq N^{-1}, \quad K_{i+1}(\tau_i - \tau_{i+1}) = \frac{hB_i}{K_i} - \delta_i, \quad B_{i+1} = B_i - (K_{i+1} + 1)\delta_i. \tag{A.2}$$

Thus $B_{i+1} \geq r(K_{i+1}/K_i)B_i > 0$ and $\beta_{N-1} > 0$.

**Lemma A.2** (Strict order)**.** *For $0 < h < 1$, the optimizer satisfies $\beta_0 > \cdots > \beta_{N-1} > 0$. In particular, $\nu = 0$.*

*Proof.* Suppose a maximal constant block $\beta_l = \cdots = \beta_b$ has $l < b$. With the convention $\nu_{-1} = 0$, complementarity at the bounding differences gives $\nu_{l-1} = \nu_b = 0$. By (A.2), $B_i = B > 0$ on the block and $\tau_i = \beta_l - B/K_i$, hence $\tau_i - \tau_{i+1} > 0$ for $l \leq i < b$, and $\mu_i = 0$ there. Put

$$e_i = \frac{\mu_i}{2hcS}, \quad f_i = \frac{\nu_i}{2hcS}, \quad m_i = (i+1)\delta_i + e_i - e_{i-1}, \quad \omega_i = m_i/K_i,$$

with $e_{-1} = e_{N-1} = f_{-1} = f_N = 0$. Let $U_0 = \tau_0$ and $U_i = (i+1)\tau_i - i\tau_{i-1}$ for $i \geq 1$. To rewrite the KKT equations, observe that

$$D^\top WT\beta = U, \qquad \omega = WD\beta + \operatorname{diag}(K_i^{-1})(e_i - e_{i-1})_{i=0}^{N-1},$$

and

$$\frac{A^\top\mu}{2hcS} = T^\top\operatorname{diag}(K_i^{-1})(e_i - e_{i-1})_{i=0}^{N-1}.$$

Substituting (3.8) into (A.1) and dividing by $2hcS$ gives

$$U + T^\top\omega + D^\top f = \kappa\mathbf{1}, \qquad \kappa = \frac{hc}{S} - \frac{\Phi_{N,h}(c)}{hcS}. \tag{A.3}$$

For $\Omega = \sum_k \omega_k$,

$$(T^\top\omega)_i = hc\Omega - h\sum_{k>i}\omega_k - \omega_i, \qquad (D^\top f)_i = f_i - f_{i-1}.$$

Subtracting coordinate $i+1$ from coordinate $i$ therefore gives

$$2f_i - f_{i-1} - f_{i+1} = \omega_i - r\omega_{i+1} - (U_i - U_{i+1}). \tag{A.4}$$

On the assumed constant block, $\omega_l \le 0$, $\omega_i = 0$ for $l < i < b$, and $\omega_b > 0$, so the first two terms on the right have nonpositive sum. Meanwhile

$$\theta_i := \frac{i+1}{K_i} - \frac{i}{K_i + h} = \frac{h(Nc+1)}{K_i(K_i+h)}$$

is strictly increasing. For $l < i \le b$, $U_i = \beta_l - B\theta_i$. At the left endpoint, $U_0 = \beta_0 - B\theta_0$ if $l = 0$, while for $l > 0$, (A.2) gives

$$\tau_{l-1} = \beta_l - \frac{B}{K_l + h} - \frac{r\delta_{l-1}}{K_l + h}, \qquad U_l = \beta_l - B\theta_l + \frac{lr\delta_{l-1}}{K_l + h} \ge \beta_l - B\theta_l.$$

Hence $U_i - U_{i+1} > 0$ for $l \le i < b$. It follows that $2f_i - f_{i-1} - f_{i+1} < 0$ there, contradicting the existence of a maximum of a nonnegative sequence $f_l, \dots, f_{b-1}$ with zero boundary values $f_{l-1} = f_b = 0$. □

By Lemma A.2, $f = 0$, so (A.3) reads $U + T^\top \omega = \kappa \mathbf{1}$. We retain $e_i \ge 0$, $e_{-1} = e_{N-1} = 0$, and $e_i(\tau_i - \tau_{i+1}) = 0$.

### A.3. **Row moments and slope blocks.**

**Lemma A.3** (Row moments and block endpoints)**.** *All $m_i$ are positive and $\sum_i m_i = 1$. If $Nh^2 \ge 1$ and $\Phi_{N,h}(c) \ge 0$, every maximal constant block $[l, b]$ of $\tau$ satisfies $m_b \ge \delta_b$, with strict inequality at a terminal block.*

*Proof.* The sum of the $m_i$ telescopes to $\sum_i (i+1)\delta_i = \sum_i \beta_i = 1$. On a maximal slope block $[l, b]$, put $g = \tau_{l-1} - \tau_l$ (zero for $l = 0$). Complementarity gives $e_{l-1} = e_b = 0$. If the block is a singleton, $m_l = (l+1)\delta_l > 0$. Otherwise the slope equality and KKT equations give

$$\delta_{i+1} = r\delta_i \ (l \le i \le b-2), \qquad \omega_i = r\omega_{i+1} \ (l < i < b), \qquad \omega_l = r\omega_{l+1} - lg. \tag{A.5}$$

Since $m_l = (l+1)\delta_l + e_l > 0$, the moment positivity propagates across the block.

For a nonterminal singleton, $m_b = (b+1)\delta_b \ge \delta_b$. For a nonterminal block of length at least two, the slope condition at its exit gives $\delta_b \le r\delta_{b-1}$; hence $\delta_i \ge \delta_b$ for $l \le i \le b$. Summing $m_i$ over the block, with $C = \omega_b$, yields

$$C \sum_{i=l}^{b} K_i r^{b-i} - K_l l g = \sum_{i=l}^{b} (i+1)\delta_i.$$

Using $K_i = K_b + h(b-i)$, $r^{b-i} \le 1$, and $g \ge 0$ gives

$$\frac{m_b}{\delta_b} \ge \frac{K_b(l+b+2)}{2K_b + h(b-l)} \ge 1. \tag{A.6}$$

The last inequality follows from $K_b = h(Nc - b) \ge 2h$ for $b \le N-2$, which implies $K_b(l+b) \ge h(b-l)$.

For the terminal block $b = N-1$, the singleton case gives $m_b = N\delta_b > \delta_b$. Otherwise write $B_* = \beta_b$, $C = \omega_b$, $\theta = \tau_b$, $\Omega = \sum_i \omega_i$. Since

$$\sum_i U_i = N\theta, \qquad \sum_i (T^\top \omega)_i = \sum_i (K_i - 1)\omega_i = 1 - \Omega,$$

summing (A.3) and then taking its last coordinate gives

$$N\kappa = N\theta + 1 - \Omega, \qquad \theta + hc\Omega - C = \kappa,$$

so $C = S\theta + \Phi_{N,h}(c)/(hc)$ and $\theta \le C/S$. For $J = N - l$, let

$$D_J = \sum_{j=0}^{J-1} (K_b + hj) r^j, \qquad H_J = h \sum_{j=1}^{J-1} (N - j) r^{-j}.$$

On the constant slope block, (A.2) gives $\beta_{i+1} - \theta = r(\beta_i - \theta)$. Therefore

$$\delta_{b-j} = hr^{-j}(B_* - \theta) \quad (1 \le j < J), \qquad \delta_b = B_*.$$

Summing the definition of $m_i$ over the block cancels the $e_i$ terms. Using (A.5) on the left and the preceding differences on the right gives

$$CD_J - K_l l g = \sum_{i=l}^{b} (i+1)\delta_i = NB_* + H_J(B_* - \theta).$$

Since $C > 0$, $g \ge 0$, and $\theta \le C/S$, this implies

$$\frac{B_*}{C} \le \frac{D_J + H_J/S}{N + H_J}. \tag{A.7}$$

We have $SK_b \ge (1 + Nh)h > 1$, and the geometric series gives

$$D_J < \sum_{j=0}^{\infty} (K_b + hj) r^j = \frac{K_b + r}{h} \le NK_b.$$

The last inequality follows from $(Nh - 1)K_b - r \ge Nh^2 - 1 \ge 0$. Hence the right side of (A.7) is strictly less than $K_b$. This proves $m_b = K_b C > B_* = \delta_b$. □

### A.4. **Construction of the directed flow.**

**Lemma A.4** (Splitting probabilities and flow)**.** *Assume $N \ge 2$, $0 < h < 1$, $Nh^2 \ge 1$, $c \ge 1$, and $\Phi_{N,h}(c) \ge 0$. For the optimizer and multipliers above, set*

$$\ell_i = \delta_i + \frac{e_i}{i+1}, \quad \pi_0 = \pi_N = 1, \quad \pi_i = 1 - \frac{e_{i-1}}{i\ell_i} \ (1 \le i < N), \quad s_i = 1 - \pi_i, \quad O_i = 1 + i\pi_i = \frac{m_i}{\ell_i}. \tag{A.8}$$

*Then $0 \le \pi_i \le 1$, and*

$$F_{ij} = O_i \pi_j \prod_{i<k<j} s_k \quad (0 \le i < j \le N) \tag{A.9}$$

*is nonnegative, satisfies (3.3), and has row moments*

$$\sum_{j>i} F_{ij}(\beta_i - \beta_j) = m_i \qquad (\beta_N = 0). \tag{A.10}$$

*Every positive edge satisfies $\tau_i = \cdots = \tau_{j-1}$.*

*Proof.* Since $e_{i-1} \ge 0$ and $\ell_i > 0$, $\pi_i \le 1$. Put $G_0 = 0$ and $G_i = im_i - e_{i-1} = i(i+1)\ell_i\pi_i$ for $1 \le i < N$. At the endpoints of a maximal slope block $[l, b]$, Lemma A.3 gives $G_l = lm_l \ge 0$ and $G_b = (b+1)(m_b - \delta_b) \ge 0$. On that block,

$$G_{i+1} - G_i = (i+1)(m_{i+1} - m_i + \delta_i).$$

Put $\chi_i = h(K_i - 1)\omega_{i+1} + \delta_i$ for $l \le i < b$. By (A.5),

$$m_{i+1} - m_i + \delta_i = \chi_i + K_l l g \, \mathbf{1}_{\{i=l\}}.$$

For $K_i \geq 1$, $\chi_i > 0$. Once $K_i < 1$, define $R_i = \delta_i/[h(1-K_i)\omega_{i+1}]$; the sign of $\chi_i$ is the sign of $R_i - 1$. For successive indices in this range, (A.5) gives

$$\frac{R_{i+1}}{R_i} = r^2 \frac{1-K_i}{1-K_i+h} < 1.$$

Thus the sequence $\chi_i$ can change sign only from positive to negative. The additional nonnegative term affects only the first increment and preserves this property. Hence $G_i$ first increases and then decreases on the block, so $G_i \geq \min\{G_l, G_b\} \geq 0$. This proves $\pi_i \geq 0$.

Since $\pi_j = 1 - s_j$ and $s_N = 0$, telescoping gives

$$\sum_{j=i+1}^{N} \pi_j \prod_{i<k<j} s_k = 1 - \prod_{k=i+1}^{N} s_k = 1.$$

Thus total outflow at $i$ is $O_i$. To compute the inflow, put $V_i = \sum_{k<i} O_k \prod_{k<j<i} s_j$. Then $V_0 = 0$ and

$$V_{i+1} = O_i + s_i V_i.$$

If $V_i = i$, the right-hand side equals $1 + i\pi_i + is_i = i + 1$. Induction gives $V_i = i$, so total inflow is $i\pi_i$ and the divergence is $O_i - i\pi_i = 1$. If $L_i$ denotes the mean gradient difference per unit outflow, then $L_{N-1} = \delta_{N-1}$ and $L_i = \delta_i + s_{i+1}L_{i+1}$. Since $s_{i+1}\ell_{i+1} = e_i/(i+1) = \ell_i - \delta_i$, backward induction gives $L_i = \ell_i$ and (A.10). Finally, if $\tau_{k-1} > \tau_k$, complementarity yields $e_{k-1} = 0$, hence $s_k = 0$. For $i < k < j$, the product defining $F_{ij}$ contains $s_k$, so $F_{ij} = 0$. Thus $F_{ij} > 0$ implies $\tau_i = \cdots = \tau_{j-1}$. □

A.5. **Positivity of the certificate matrix.**

**Proposition A.5.** *Under the hypotheses of Proposition 3.2, the flow* (A.9) *satisfies* $M(F) \succeq 0$.

*Proof.* Define a lower triangular matrix $C$ by $C_{ii} = O_i/K_i$ and $C_{ji} = -F_{ij}/K_i$ for $i < j < N$. The row-moment identity gives $C^\top \beta = \omega$. The support property of the flow gives $CT\beta = U$; indeed, for $i \geq 1$,

$$(CT\beta)_i = O_i\tau_i - \sum_{k<i} F_{ki}\tau_k = (1+i\pi_i)\tau_i - i\pi_i\tau_{i-1} = U_i,$$

where $(1-\pi_i)(\tau_{i-1} - \tau_i) = 0$. The matrix satisfies $M(F) = h^2c^2\mathbf{1}\mathbf{1}^\top - hcS(CT + T^\top C^\top)$, so the KKT equation yields

$$M(F)\beta = \Phi_{N,h}(c)\mathbf{1}. \tag{A.11}$$

It remains to determine the signs of the off-diagonal entries. Put

$$\alpha_i = \frac{O_i}{K_i} = \frac{\omega_i}{\ell_i}, \quad Z_0 = 0, \quad Z_{i+1} = \alpha_i + s_i Z_i, \quad X_i = \frac{s_i Z_i}{\alpha_i}.$$

At the beginning of each maximal slope block, $s_i = X_i = 0$. Within a block,

$$X_{i+1} = \frac{\omega_i}{\omega_{i+1}}\left(1 - \frac{\delta_i}{\ell_i}\right)(1 + X_i) < r(1 + X_i),$$

and at a block boundary $X_{i+1} = 0$. Therefore $0 \leq X_i < r/h$ and $r\alpha_i - hs_iZ_i > 0$. Let $\rho_i = \alpha_i - \pi_i Z_i$. Flow conservation and $K_k > K_i$ for $k < i$ imply

$$\rho_i = \frac{\mathrm{out}_i}{K_i} - \sum_{k<i} \frac{F_{ki}}{K_k} \geq \frac{1}{K_i}.$$

To compute the off-diagonal entries, note that

$$Z_i = \sum_{k<i} \alpha_k \prod_{k<r<i} s_r.$$

For $i<j$, set $P_{ij}=\pi_j\prod_{i<k<j}s_k$. The product formula for the flow gives

$$\frac{F_{ij}}{K_i}=P_{ij}\alpha_i,\qquad \sum_{k\le i}\frac{F_{kj}}{K_k}=P_{ij}(\alpha_i+s_iZ_i).$$

Using $T_{ki}=h(c-1)+h\mathbf{1}_{\{k\le i\}}-\mathbf{1}_{\{k=i\}}$, we obtain

$$\begin{aligned}(CT)_{ij}&=hc\rho_i,\\ (CT)_{ji}&=h(c-1)\rho_j-h\sum_{k\le i}\frac{F_{kj}}{K_k}+\frac{F_{ij}}{K_i}\\ &=h(c-1)\rho_j+P_{ij}(r\alpha_i-hs_iZ_i).\end{aligned}$$

Substitution into the expression for $M(F)$ gives

$$M(F)_{ij}=h^2c^2-hcS\left[hc\rho_i+h(c-1)\rho_j+\pi_j\prod_{i<k<j}s_k\,(r\alpha_i-hs_iZ_i)\right]<0, \tag{A.12}$$

since the bracket is at least $hc/K_i>hc/S$. We have $\beta_i>0$ by Lemma A.2. Together with (A.11), this gives the exact decomposition, for every $z\in\mathbb{R}^N$,

$$z^\top M(F)z=\sum_{i<j}(-M(F)_{ij})\beta_i\beta_j\left(\frac{z_i}{\beta_i}-\frac{z_j}{\beta_j}\right)^2+\Phi_{N,h}(c)\sum_i\frac{z_i^2}{\beta_i}\ge 0. \tag{A.13}$$

Thus $M(F)\succeq 0$. □

## Appendix B. Tridiagonal and endpoint estimates

### B.1. **Energy reduction and positivity.**

**Lemma B.1.** *Equations (4.4)–(4.6) hold.*

*Proof.* By (4.2),

$$\frac{w_i-tp_i}{d_i}=z_i-\frac{t(\zeta_i-1)}{d_i}u_i,\quad \sum_i(i+1)(p_i-p_{i+1})z_i=\mathfrak{F}\sum_i u_iz_i-\sum_{i\ge1}i(\zeta_i-1)u_{i-1}u_i.$$

Using $2\sum_i u_iz_i=(\sum_i u_i)^2+\sum_i u_i^2$, the squared sum cancels the first term of (4.3). The remaining diagonal and adjacent coefficients are (4.5), using $1-t(\zeta_i-1)/d_i=-(t-1)\zeta_i/d_i$ for the latter.

For positivity write

$$E=\operatorname{diag}(e_i(\mathfrak{F}))+\frac{t-1}{\mathfrak{F}}H,\qquad e_i(\mathfrak{F})=2i+1-\frac{2i(i+1)}{\mathfrak{F}},$$

where $H$ is symmetric tridiagonal and

$$H_{ii}=\frac{2(i+1)\zeta_i^2}{d_i},\qquad H_{i,i+1}=-\frac{(i+1)\zeta_i(\zeta_i-2)}{d_i}.$$

Its off-diagonal entries are nonpositive. The sum of row $i$ is at least

$$\frac{(i+1)\zeta_i(\zeta_i+2)}{d_i}-\frac{i(\zeta_i^2-1)}{d_i+1}>\frac{\zeta_i^2+2(i+1)\zeta_i+i}{d_i+1}>0.$$

At $i=N-1$, the absent right off-diagonal entry contributes zero to the row sum. Writing the quadratic form as a sum of nonnegative weighted adjacent differences and strictly positive row-sum squares shows $H\succ 0$. Since $\mathfrak{F}\ge N+1$ and $e_i(\mathfrak{F})\ge e_i^0>0$, (4.6) follows. □

B.2. **Strict feasibility of the trial vector.**

**Lemma B.2.** *The trial vector in Lemma 4.3 satisfies both strict orders for* $0 \le a \le (2t+1)^{-1}$.

*Proof.* Write $s = 2t$. The differences of $v_i$ in (4.14) are positive, including $v_{N-1} > 0$; hence $D\beta > 0$. For $j = N - i \ge 2$, clearing the positive denominators in the slope inequality gives

$$\mathscr{L}_i(v) := (j + a + t)v_i - \sum_{k=i}^{N-1} v_k - t(j+a)(v_i - v_{i+1}) > 0. \tag{B.1}$$

We check the two summands of $v$ separately. For $v_i^L = (i+s)^{-1}$, put $u = i + s \ge 2t$. Since

$$u \sum_{k=i}^{N-1} \frac{1}{k+s} = j - \sum_{k=0}^{j-1} \frac{k}{u+k}, \qquad \sum_{k=0}^{j-1} \frac{k}{u+k} \ge \frac{j(j-1)}{2(u+j-1)},$$

one obtains

$$u\mathscr{L}_i(v^L) \ge a\left(1 - \frac{t}{u+1}\right) + t - \frac{tj}{u+1} + \frac{j(j-1)}{2(u+j-1)} > 0.$$

The last three terms have numerator $2tu^2 + (j-1)[j(u+1-2t) + 2t] > 0$ after multiplication by $2(u+1)(u+j-1)$; the first is nonnegative.

For $v_i^R = q_j^{\mathrm{H}}/N$, summation gives $\sum_{k=1}^{j} q_k^{\mathrm{H}} = (j+s+1)q_j^{\mathrm{H}} - j$, and hence

$$N\mathscr{L}_i(v^R) = j + (a - t - 1)q_j^{\mathrm{H}} - \frac{t(j+a)}{j+2t}.$$

Using $q_j^{\mathrm{H}} \le j/(2t+1)$ and $a \le (2t+1)^{-1}$ gives

$$N\mathscr{L}_i(v^R) \ge \frac{t}{j+2t}\left(\frac{j(j-1)}{2t+1} - a\right) \ge \frac{t[j(j-1)-1]}{(j+2t)(2t+1)} > 0.$$

Thus (B.1) holds. The shifted slope formula gives

$$\tau_i - \tau_{i+1} = \frac{m\,\mathscr{L}_i(v)}{H_N(j+a)(j+a-1)} > 0 \qquad (0 \le i < N-1),$$

so $A\beta > 0$. □

B.3. **Reference energy and parameter estimates.** The reference energy admits an explicit reciprocal model. The following comparison controls its error over all indices and reduces the leading term to two harmonic sums.

**Lemma B.3** (Reference energy)**.** *For fixed* $t \ge 1$*, the vector* $v$ *in* (4.14) *satisfies*

$$\mathcal{E}_{0,1}(v) = 4\log N + O_t(1). \tag{B.2}$$

*Proof.* Keep $s = 2t$ fixed in $v$, and put $j = N - i$, $\xi_i = i + s$, $\Lambda = N + s$, and $L_i = \sum_{k=i}^{N-1}(k+s)^{-1}$. At energy parameters $(a,t) = (0,1)$, the transformed coordinate is

$$u_i^0 = \frac{(j+1)/\xi_i - L_i + (j - sq_j^{\mathrm{H}})/N}{j(j+1)}. \tag{B.3}$$

Since the reference matrix is diagonal,

$$\mathcal{E}_{0,1}(v) = (N+1)^2 \sum_i e_i^0 (u_i^0)^2. \tag{B.4}$$

We compare $u_i^0$ with $\widehat{u}_i = (j\xi_i)^{-1}$. Write

$$r_i = L_i - \frac{j}{\Lambda} = \sum_{k=i}^{N-1} \frac{N-k}{\Lambda(k+s)} \geq 0.$$

As $j + \xi_i = \Lambda$, subtraction in (B.3) gives

$$u_i^0 - \widehat{u}_i = \frac{s}{N\Lambda(j+1)} - \frac{r_i}{j(j+1)} - \frac{sq_j^{\mathrm{H}}}{Nj(j+1)}. \tag{B.5}$$

Each of the three terms has summable absolute value. For the middle term, reversing the order of summation gives

$$\begin{aligned}\sum_i \frac{r_i}{j(j+1)} &= \sum_{k=0}^{N-1} \frac{N-k}{\Lambda(k+s)} \sum_{i=0}^{k} \frac{1}{(N-i)(N-i+1)} \\ &= \frac{1}{\Lambda(N+1)} \sum_{k=0}^{N-1} \frac{k+1}{k+s} \leq \frac{1}{N}.\end{aligned}$$

For the last term,

$$\sum_{j=1}^{N} \frac{q_j^{\mathrm{H}}}{j(j+1)} = \sum_{k=1}^{N} \frac{1}{k+s} \left( \frac{1}{k} - \frac{1}{N+1} \right) \leq \sum_{k=1}^{N} \frac{1}{k(k+1)} \leq 1.$$

The first term sums to at most $s/N$, since $\sum_{j=1}^{N} (j+1)^{-1} \leq N$ and $\Lambda \geq N$. Consequently,

$$\sum_i |u_i^0 - \widehat{u}_i| \leq \frac{1+2s}{N}. \tag{B.6}$$

To pass from this bound to the energy, note that $\xi_i^{-1} \leq L_i \leq j/\xi_i$ and $0 < j - sq_j^{\mathrm{H}} \leq j$. Equation (B.3) therefore gives

$$0 < u_i^0 \leq \widehat{u}_i + \frac{1}{N(j+1)}, \qquad u_i^0 + \widehat{u}_i \leq (2+s)\widehat{u}_i,$$

where the second inequality uses $\xi_i \leq Ns$. Also, $\xi_i j \geq N$ and $e_i^0 = 1 + 2ij/(N+1)$ imply

$$e_i^0 \leq \frac{3\xi_i j}{N}, \qquad (N+1)^2 e_i^0 (u_i^0 + \widehat{u}_i) \leq 12(2+s)N. \tag{B.7}$$

Define $\mathcal{M}_N = (N+1)^2 \sum_i e_i^0 \widehat{u}_i^2$. Factoring the difference of squares and using (B.6) yields

$$|\mathcal{E}_{0,1}(v) - \mathcal{M}_N| \leq 12(2+s)N \sum_i |u_i^0 - \widehat{u}_i| \leq 12(2+s)(1+2s). \tag{B.8}$$

It remains to sum the model. Partial fractions give

$$\frac{(N+1)^2 e_i^0}{j^2 \xi_i^2} = \mathsf{A}_N \left( \frac{1}{\xi_i} + \frac{1}{j} \right) + \frac{\mathsf{B}_N}{\xi_i^2} + \frac{\mathsf{D}_N}{j^2}, \tag{B.9}$$

where

$$\mathsf{A}_N = \frac{2N(N+1)}{\Lambda^2} + \frac{2(N+1)^2}{\Lambda^3} = 2 + O_t(N^{-1}),$$
$$\mathsf{B}_N = \frac{(N+1)^2}{\Lambda^2} - \frac{2s(N+1)}{\Lambda} = O_t(1), \qquad \mathsf{D}_N = \frac{(N+1)^2}{\Lambda^2} = O_t(1).$$

Both reciprocal sums in (B.9) equal $\log N + O_t(1)$, while the squared reciprocal sums are bounded. Thus $\mathcal{M}_N = 4 \log N + O_t(1)$, and (B.8) proves (B.2). □

**Lemma B.4.** *For fixed $t \geq 1$, the estimates (4.15) hold uniformly for $0 \leq a \leq (2t+1)^{-1}$.*

*Proof.* Interchanging summations yields

$$H_N = \sum_{i=0}^{N-1} \frac{1}{i+s} + \frac{N+s+1}{N} q_N^{\mathrm{H}} - 1 = 2\log N + O_t(1). \tag{B.10}$$

Lemma B.3 evaluates the unnormalized energy at $(a,t) = (0,1)$. We now estimate how this energy changes over the prescribed range of $a$; the vector $v$, with $s = 2t$, is held fixed throughout this comparison.

Put $\gamma_i = (i+1)(v_i - v_{i+1})$, $W_i = \sum_{k=i}^{N-1} v_k$, and $J_{a,t}(v) = \sum_i \gamma_i (W_i - t v_i)/(N+a-i)$. Then

$$J_{a,t}(v) - J_{0,1}(v) = -\sum_i \gamma_i \left[ \frac{a(W_i - v_i)}{j(j+a)} + \frac{(t-1)v_i}{j+a} \right], \qquad j = N - i. \tag{B.11}$$

Using $0 \leq W_i - v_i \leq (j-1)v_i$ gives

$$|J_{a,t}(v) - J_{0,1}(v)| \leq (a+t-1) \sum_i \frac{\gamma_i v_i}{j} = O_t(N^{-1}).$$

For the last estimate, split at $i = N/2$. For $i \leq N/2$, use

$$v_i \leq C_t \left( (i+1)^{-1} + \frac{\log N}{N} \right), \quad \gamma_i \leq C_t \left( (i+1)^{-1} + \frac{i+1}{N^2} \right), \quad N/j \leq 2.$$

Multiplying these bounds and summing gives

$$N \sum_{i \leq N/2} \frac{\gamma_i v_i}{j} \leq C_t \left( \sum_{i \geq 0} \frac{1}{(i+1)^2} + \frac{\log^2 N}{N} + 1 \right) = O_t(1).$$

For $i > N/2$ and $j \geq 2$, use $v_i \leq C_t(1 + \log j)/N$ and $\gamma_i \leq C_t(N^{-1} + j^{-1})$. Then

$$N \sum_{\substack{i > N/2 \\ j \geq 2}} \frac{\gamma_i v_i}{j} \leq C_t \sum_{j=2}^{N} \left( \frac{1 + \log j}{Nj} + \frac{1 + \log j}{j^2} \right) = O_t(1).$$

At $j = 1$, $v_{N-1} = O_t(N^{-1})$ and $\gamma_{N-1} = N v_{N-1} = O_t(1)$ give the same estimate.

Since $H_N^2 - 2(N+1)J_{0,1}(v) = \mathcal{E}_{0,1}(v)$, we have $J_{0,1}(v) = O_t(\log^2 N/N)$. Therefore

$$\mathcal{E}_{a,t}(v) - \mathcal{E}_{0,1}(v) = -2(a+t-1)J_{0,1}(v) - 2\mathfrak{F}[J_{a,t}(v) - J_{0,1}(v)] = O_t(1) + O_t(\log^2 N/N),$$

uniformly in $a \in [0, (2t+1)^{-1}]$. Dividing by $H_N^2$ and using (B.10), (B.2) establishes (4.15). □

## Verification and code

The accompanying Python programs check finite instances of the interpolation identities, KKT systems, flow certificates, energy formulas, and planar counterexamples using exact rational arithmetic. The checks require only the Python standard library and can be run from the project root with `python scripts/verify.py`. The source distribution includes their input data and verification records.

The main theorem and its supporting results are formalized in Lean 4.26.0, with the mathlib revision and transitive dependencies fixed by the project manifest. The complete build and axiom audit cover 699 theorem declarations, including auxiliary lemmas; their axiom dependencies are `propext`, `Classical.choice`, and `Quot.sound`. From the `lean_formalization/` directory, `lake build` compiles the development and `pwsh -File check_complete.ps1` runs the complete audit. The source distribution also includes a proof map and source-file hashes. The global comparison in Lemma B.3 was developed during the formalization.

## AI assistance

AI tools, including OpenAI Codex and other AI assistants, were used to explore proof strategies, perform algebraic and numerical calculations, develop verification code, assist the Lean formalization, and revise the exposition. The author is responsible for the mathematical statements, proofs, and final text.

## References

[1] Y. Drori and M. Teboulle, *Performance of first-order methods for smooth convex minimization: A novel approach*, Math. Program. **145** (2014), 451–482.

[2] A. Luner and B. Grimmer, *On averaging and extrapolation for gradient descent*, Math. Oper. Res. **51** (2026), no. 3, 2158–2188. DOI: 10.1287/moor.2024.0449. See also arXiv:2402.12493v4, Conjecture 5.1.

[3] A. B. Taylor, J. M. Hendrickx, and F. Glineur, *Smooth strongly convex interpolation and exact worst-case performance of first-order methods*, Math. Program. **161** (2017), 307–345.

[4] M. Teboulle and Y. Vaisbourd, *An elementary approach to tight worst case complexity analysis of gradient based methods*, Math. Program. **201** (2023), 63–96.